```
Dataset shape: (50, 17)

First 5 rows:
   duration protocol_type service flag  src_bytes  dst_bytes  count  \
0         0           tcp    http   SF        181       5450      8
1         0           tcp    http   SF        239        486      8
2         0           tcp    http   SF        235       1337      8
3         0           tcp    http   SF        219       1337      6
4         0           tcp    http   SF        217       2032      6

   srv_count  serror_rate  srv_serror_rate  same_srv_rate  diff_srv_rate  \
0          8          0.0              0.0            1.0            0.0
1          8          0.0              0.0            1.0            0.0
2          8          0.0              0.0            1.0            0.0
3          6          0.0              0.0            1.0            0.0
4          6          0.0              0.0            1.0            0.0

   dst_host_count  dst_host_srv_count  dst_host_same_srv_rate  \
0               9                   9                     1.0
1              19                  19                     1.0
2              29                  29                     1.0
3              39                  39                     1.0
4              49                  49                     1.0

   dst_host_serror_rate   label
0                   0.0  normal
1                   0.0  normal
2                   0.0  normal
3                   0.0  normal
4                   0.0  normal

================ MODEL COMPARISON ================
Decision Tree Accuracy : 0.93
Random Forest Accuracy : 1.00

Decision Tree Confusion Matrix:
[[6 1]
 [0 8]]

Random Forest Confusion Matrix:
[[7 0]
 [0 8]]

Decision Tree Classification Report:
              precision    recall  f1-score   support

      attack       1.00      0.86      0.92         7
      normal       0.89      1.00      0.94         8

    accuracy                           0.93        15
   macro avg       0.94      0.93      0.93        15
weighted avg       0.94      0.93      0.93        15


Random Forest Classification Report:
              precision    recall  f1-score   support

      attack       1.00      1.00      1.00         7
      normal       1.00      1.00      1.00         8
```

```
              accuracy                           1.00        15
             macro avg       1.00      1.00      1.00        15
          weighted avg       1.00      1.00      1.00        15
```

Conclusion: Random Forest performed better than Decision Tree.

In [ ]: